⭐ System Design



➡️ 1. System Design Key Concepts - Video Lectures

Sr No	Topic	Link
1	API Design	YT
2	API Gateway	YT
3	Json Web Token	YT
4	Webhooks	YT
5	Load Balancing	YT
6	Proxy vs Rev Proxy	YT
7	CAP Theorem	YT
8	Sharding	YT
9	Caching	YT
10	Scaling	YT
11	Availability	YT

12	Services	YT
13	Databases	YT
14	Data Sharding & Partitioning	YT
15	REST	YT
16	Distributed Systems	YT
17	SQL vs No SQL	YT
18	ACID Transactions	YT
19	Consistent Hashing	YT
20	CDC	YT
21	Caching	YT
22	Caching Strategies	YT
23	Cache Eviction	YT
24	CDN	YT
25	Rate Limiting	YT
26	Message Queues	YT
27	Bloom Filters	YT
28	Idompotency	YT
29	Concurrency vs Parallelism	YT
30	Long Polling vs Websockets	YT
31	Stateful vs Stateless	YT
32	Batch vs Stream	YT
33	Sync vs Async	YT

2. System Design Key Concepts - Blogs

1. Scalability: https://lnkd.in/gpge_z76
2. Latency vs Throughput: https://lnkd.in/g_amhAtN
3. CAP: https://lnkd.in/g3hmVamx
4. ACID Transactions: https://lnkd.in/gMe2JqaF
5. Rate Limiting: https://lnkd.in/gWsTDR3m
6. API Design: https://lnkd.in/ghYzrr8q

7. Strong vs Eventual Consistency: https://lnkd.in/gJ-uXQXZ
8. Distributed Tracing: https://lnkd.in/d6r5RdXG
9. Synchronous vs. asynchronous communications: https://lnkd.in/gC3F2nvr
10. Batch Processing vs Stream Processing: https://lnkd.in/g4_MzM4s
11. Fault Tolerance: https://lnkd.in/dVJ6n3wA

➤ System Design Building Blocks:

1. Database: https://lnkd.in/gti8gjpz
2. Horizontal vs Vertical Scaling: https://lnkd.in/gAH2e9du
3. Caching: https://lnkd.in/gC9piQbJ
4. Distributed Caching: https://lnkd.in/g7WKydNg
5. Load Balancing: https://lnkd.in/gQaa8sXK
6. SQL vs NoSQL: https://lnkd.in/g3WC_yxn
7. Database Scaling: https://lnkd.in/gAXpSyWQ
8. Data Replication: https://lnkd.in/gVAJxTpS
9. Data Redundancy: https://lnkd.in/gNN7TF7n
10. Database Sharding: https://lnkd.in/gMqqc6x9
11. Database Index's: https://lnkd.in/gCeshYVt
12. Proxy Server: https://lnkd.in/gi8KnKS6
13. WebSocket: https://lnkd.in/g76Gv2KQ
14. API Gateway: https://lnkd.in/gnsJGJaM
15. Message Queues: https://lnkd.in/gTzY6uk8

➤ System Design Architectural Patterns:

1. Event-Driven Architecture: https://lnkd.in/dp8CPvey
2. Client-Server Architecture: https://lnkd.in/dAARQYzq
3. Serverless Architecture: https://lnkd.in/gQNAXKkb
4. Microservices Architecture: https://lnkd.in/gFXUrz_T

➤ Low-Level Design Problems:

1. Design Parking Lot: https://lnkd.in/dQaAuFd2
2. Design Splitwise: https://lnkd.in/dF5fBnex
3. Design Chess Validator: https://lnkd.in/dfAQHvN4
4. Design Distributed Queue | Kafka: https://lnkd.in/dQ6_B4_M

➤ System Design and Architecture (HLD):

1. Design Unique ID Generator Service
2. Design bitly
3. Design Whatsapp
4. Design Insta/Twitter News Feed
5. Design Search Autocomplete